WEATHER MLOPS · DATASCIENTEST MLOPS BOOTCAMP JUL26

Monitoring dashboards

Grafana screenshots of the running system

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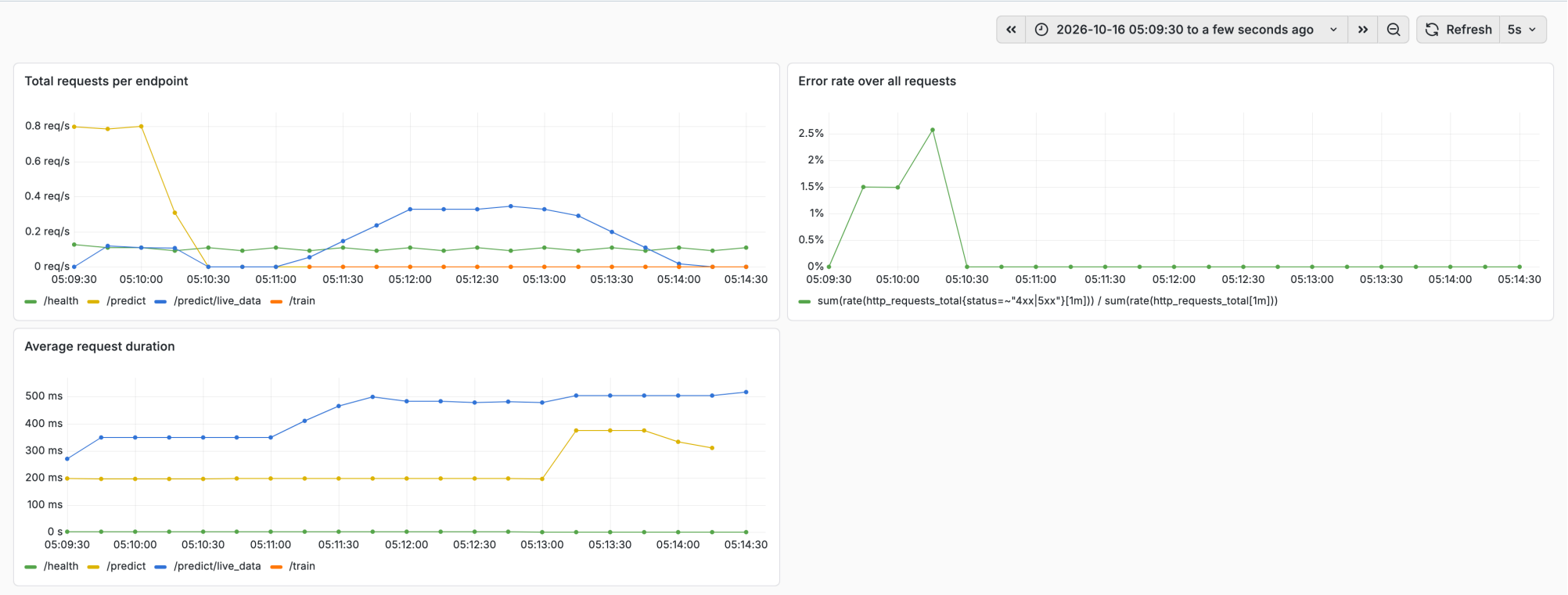
Mentor: Nicolas Fradin

Captured on 16 October 2026 from `make streamlit` (Prometheus, Pushgateway, Grafana, Airflow, MLflow, API behind Nginx)

Code: commit 2063257

FIGURE 1

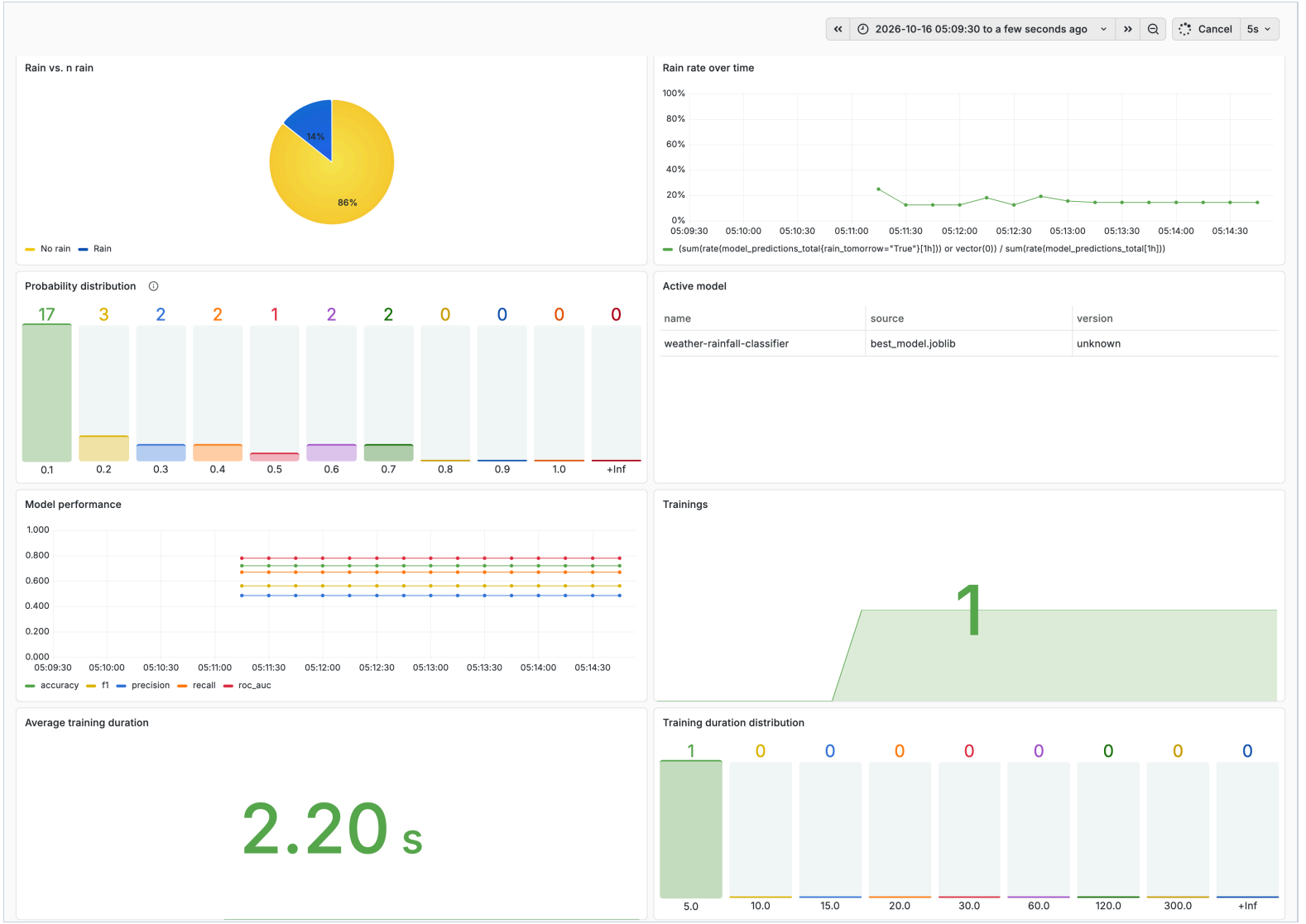
Weather API dashboard



Traffic through the Nginx gateway to FastAPI: requests per endpoint (`/health` , `/predict` , `/predict/live_data` , `/train`), error rate over all requests and average latency. The small error bump (2.5 %) is one deliberate request without credentials (HTTP 401). Prometheus scrapes the API every 5 s.

FIGURE 2

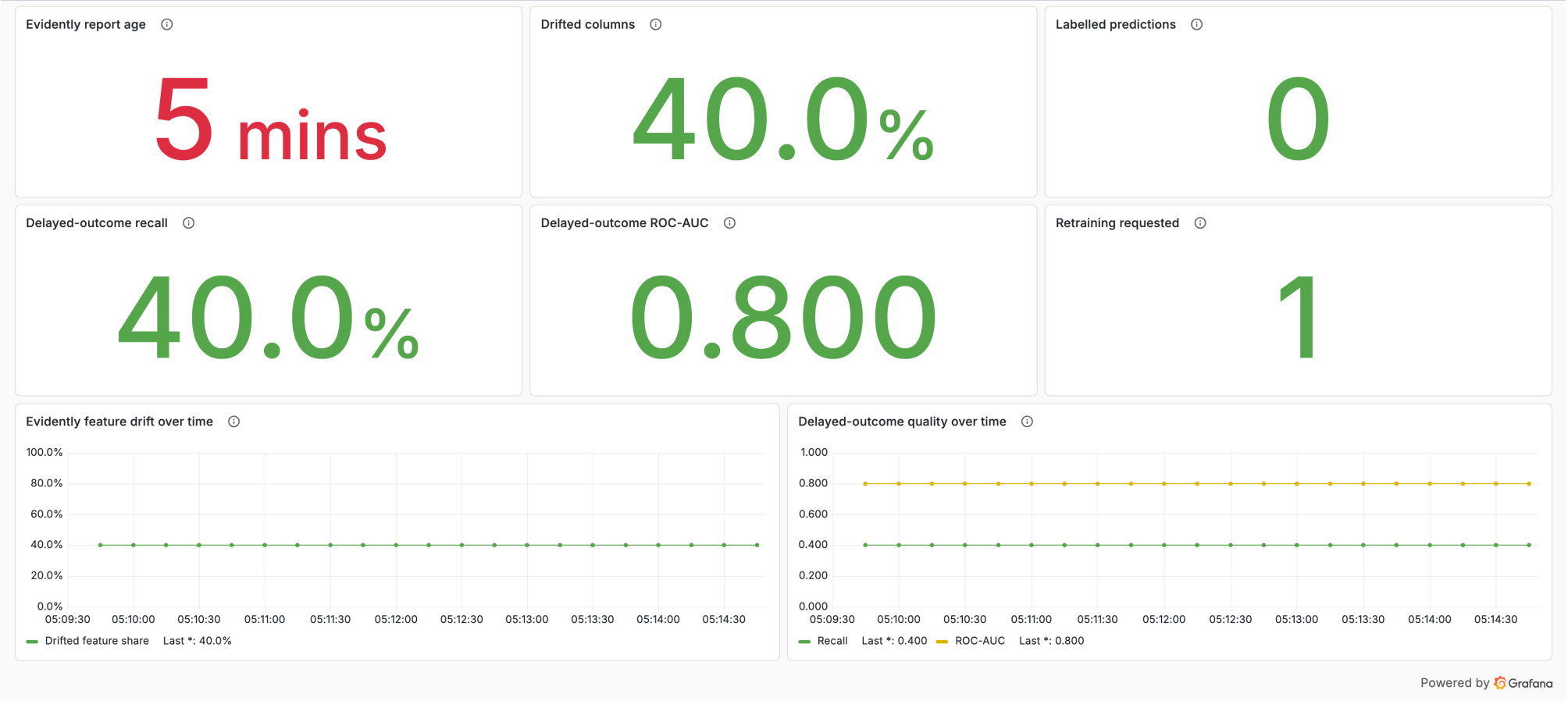
Weather Model dashboard (part 1): predictions and training



Prediction mix (14 % rain), probability histogram and rain rate of the live predictions in the time window shown; the served model; and the metrics, count and duration of the last training. The served model is the restored local copy `best_model.joblib`, because this machine's MLflow held no `@champion` alias; the training was run through the API without MLflow tracking, so it cannot move `@champion`.

FIGURE 3

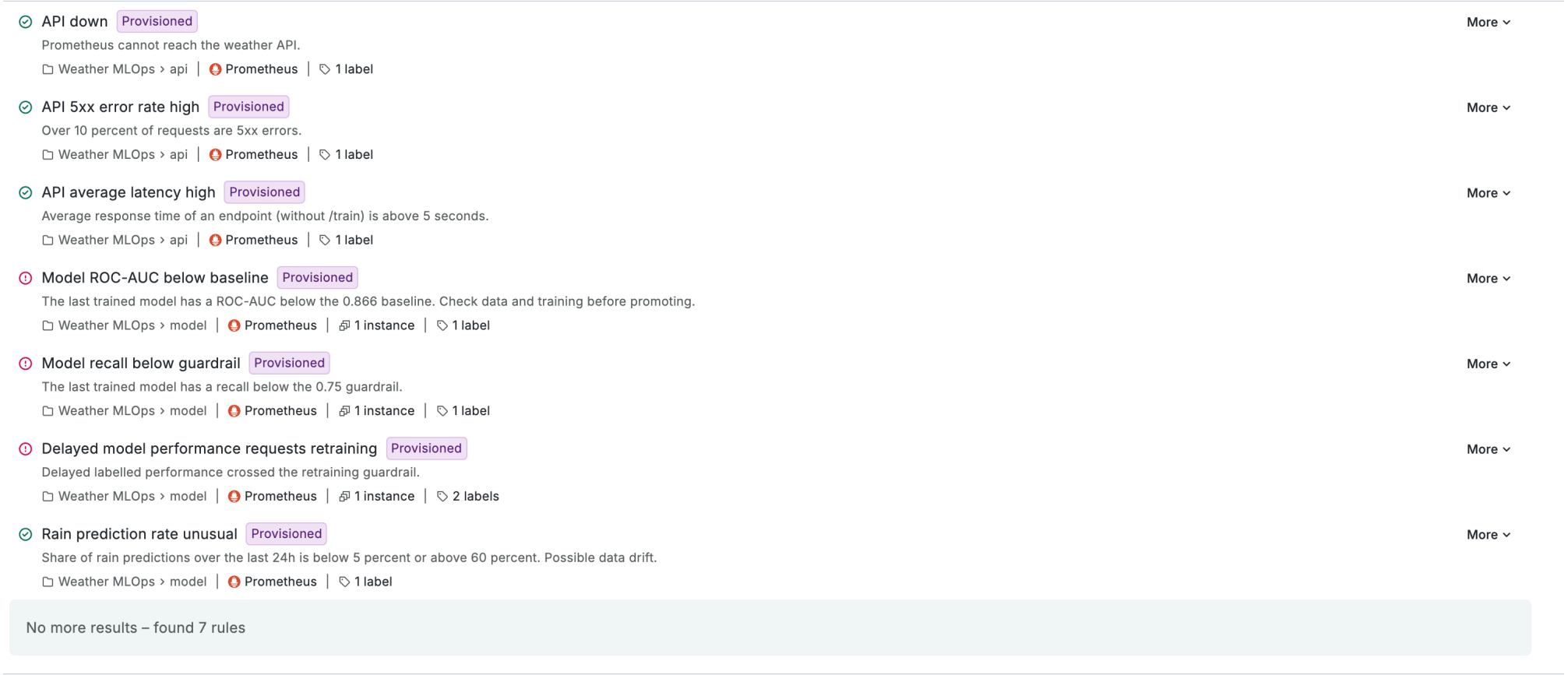
Weather Model dashboard (part 2): drift and delayed quality



Evidently feature drift (40 % of the five monitored features) and delayed-outcome quality (recall 40 %, ROC-AUC 0.800). These values come from the defense simulation `POST /monitor/simulate`, which injects a feature shift and a preset recall instead of real outcomes; no prediction has a known outcome yet, hence *Labelled predictions = 0*. The simulated recall is below the 0.75 guardrail, so *Retraining requested = 1*.

FIGURE 4

Grafana alert rules



The seven provisioned alert rules. Firing: the last training is below the 0.866 ROC-AUC baseline and the 0.75 recall guardrail, and delayed performance requests retraining, which calls the Airflow `weather_retrain` DAG through its REST API (at most once every 24 h). The three API rules and the rain-rate rule are normal.